Tianxin Hu

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Educational Background

Nanyang Technological University (Master of Science) **Aug 2024 – Jan 2026**

- School: Electrical and Electronic Engineering
- Major: Computer Control & Automation
- GPA: 95.00/100

Wuhan University of Science and Technology (Bachelor of Engineering) **Sept 2020 – Jun 2024**

- School: Automobile and Traffic Engineering
- Major: Vehicle Engineering (Industrial Planning)
- GPA: 88.30/100 (Class Ranking: 3/32 Department Ranking: 8/157)

Research & Academic Experience

Chery Robotics — Humanoid Robot Locomotion Development **Mar 2026 – Present**

- **Supervisor:** Professor Lihua Xie
- **Position:** Research Assistant
- **Project:** Industry-oriented humanoid robotics project targeting language-guided embodied task execution, such as object fetching in real-world environments
- **Responsibilities:** Responsible for the locomotion module of Chery's humanoid robot, with a focus on reinforcement learning-based humanoid locomotion training and deployment

AIBOT Technology Pte. Ltd. — Autonomous Navigation System Development **Jul 2025 – Present**

- **Supervisor:** Professor Lihua Xie
- **Position:** Navigation Module Lead
- **Responsibilities:** Developed the full-stack navigation module for the company's autonomous mobile robot, responsible for cross-floor navigation, elevator interface integration, and dynamic obstacle avoidance within residential and commercial environments

✓ **Achievements:** Achieved fully autonomous navigation and delivery within multi-story residential complexes

Research Student — NTU Internet of Things Lab & Center for Advanced Robotics Technology and Innovation (CARTIN) **Aug 2024 – Mar 2026**

- **Supervisors:** Professor Lihua Xie
- **Research Focus:** Developed optimization-based motion planning methods for multi-axle autonomous vehicles to minimize swept volume during turning maneuvers and reduce tire wear under dynamic constraints

✓ **Achievements:** Published five papers on vehicle planning and control (three under review) [J1-J3]

Xin Chi Technology — Research and Development of Autonomous Driving System for Multi-Axle Heavy-Duty Transport Vehicles **Jun 2022 – Jun 2024**

- **Supervisor:** Senior Engineer Xiangrong You
- **Position:** Technical Director & Corporate Legal Representative
- **Responsibilities:** Led R&D of autonomous driving systems for heavy-duty vehicles, covering mathematical modeling, ROS architecture, microcontroller development, and both simulation and prototype validation

✓ **Achievements:** Contributed to multiple patents [P1–P4] and received several national awards [A1–A5]

Research and Development of Autonomous Driving System for Formula Racing **Feb 2021 – Jun 2024**

- **Supervisors:** Associate Professor Chao Deng, Dr. Junyi Zou
- **Position:** Leader of the Unmanned Group
- **Responsibilities:** Led the development of the autonomous driving system for Formula Student Autonomous China (FSAC), covering visual detection, mapping, planning, and tracking

✓ **Achievements:** Received the FSAC Third Prize and Best Rookie Award [A6], and published a paper on it [J4]

Publications

[J1] **Swept Volume-Aware Trajectory Planning and MPC Tracking for Multi-Axle Swerve-Drive AMRs** **May 2025**
IEEE International Conference on Robotics and Automation ***(ICRA 2025)***
Tianxin Hu, Shenghai Yuan, Ruofei Bai, Xinhang Xu, Yuwen Liao, Fen Liu, Lihua Xie
DOI: 10.1109/ICRA55743.2025.11127914

[J2] **Tire Wear-Aware Trajectory Tracking Control for Multi-Axle Swerve-Drive Autonomous Mobile Robots** **Jun 2025**
Journal of Automation and Intelligence ***(JAI)***
Tianxin Hu, Xinhang Xu, Thien-Minh Nguyen, Fen Liu, Shenghai Yuan, Lihua Xie
DOI: 10.1016/j.jai.2025.05.003

[J3] **TGSFormer: Scalable Temporal Gaussian Splatting for Embodied Semantic Scene Completion** **Mar 2026**
IEEE Conference on Computer Vision and Pattern Recognition ***(CVPR 2026)***
Rui Qian, Haozhi Cao, Tianchen Deng, **Tianxin Hu**, Weixing Guo, Shenghai Yuan, Lihua Xie
DOI: 10.48550/arXiv.2512.00300

[J4] **Improvement of YOLOv5 Algorithm for Formula Racing Based on TensorRT under ROS Platform** **May 2023**
Agricultural Equipment & Vehicle Engineering
Tianxin Hu, Chao Deng, Junjie Ma, Wang Liu
DOI: 10.3969/j.issn.1673-3142.2023.05.004

Patents

[P1] **(Granted) Multi-axle all-wheel steering vehicle vector-division correction method and system** **Sep 2025**
Tianxian Hu, Xiangrong You, Dongmei Wang, Xinhe Long, Xinyu Wang, Shuang Liu, Yutong Fu, Xinyao Wang, Lefeng Wei, Baizhen Zhang
Patent No.: CN116534005B

[P2] **(Granted) A system and method for automatically correcting the deviation of a beam transport vehicle in a tunnel** **Jul 2023**
Xiangrong You, Enze Jin, Dongmei Wang, Yuran Huang, **Tianxian Hu**, Shuai Zhu, Beijia Zhang
Patent No.: CN114633801B

[P3] **(Pending) Energy-saving system and method for beam transport vehicle based on traveling pressure control** **Feb 2024**
Wei Li, Xiangrong You, Dongmei Wang, Xinhe Wang, Peng Cheng, Yuming Liu, Shuang Hu, **Tianxian Hu**, Xinyao Wang, Lefeng Wei, Baizhen Zhang
Application No.: CN117508238A

[P4] **(Pending) Method, system and equipment for planning obstacle avoidance path of multi-shaft all-wheel steering beam transporting vehicle** **Dec 2023**
Xiangrong You, Yucheng Yin, Liwei Wang, Dongmei Wang, **Tianxian Hu**, Shuang Liu, Xinyao Wang, Lefeng Wei, Baizhen Zhang
Application No.: CN117168481A

Awards

[A1] **(International Level) Bronze Award of the 9th China International College Students Innovation Competition** **Apr 2024**
Organizing Committee of China College Students "Internet+" Competition

[A2] **(National Level) The First Prize of the 18th "Challenge Cup" National Undergraduate Curricular Academic Science and Technology Works** **Oct 2023**
China Association for Science and Technology, Ministry of Education of the PRC, Chinese Academy of Social Sciences

[A3] **(International Level) Bronze Award of the 8th China College Students "Internet+" Innovation and Entrepreneurship Competition** **Apr 2023**
Organizing Committee of China College Students "Internet+" Competition

[A4] **(National Level) Second Prize of the 15th China College Student Computer Design Competition** **Jul 2022**

Organizing Committee of Chinese College Student Computer Design Competition

[A5] (National Level) Second Prize of the 15th Energy Saving and Emission Reduction Competition **Aug 2022**
China College Student Energy Conservation and Emission Reduction Social Practice and Technology Competition Committee

[A6] (National Level) Third Prize and Best Rookie Award of Formula Student Competition **Oct 2022**
China Society of Automotive Engineers, Organizing Committee of FSAC

Technical Expertise

Programming: C, C++, Python, XML (URDF/Launch)
Robotics & Simulation: ROS/ROS2, Isaac Sim/Lab, Gazebo Classic & GZ Sim, RViz, URDF/SDF modeling
Embedded Systems: NVIDIA Jetson Series, STM32, ESP8266, 8051; FreeRTOS, LVGL, U8G2
Hardware Design: Altium Designer, Proteus; multilayer PCB design and soldering
Modeling, Simulation & CAD: MATLAB, COMSOL, SolidWorks, CATIA, Blender, AutoCAD